

Customer Sign-Up Behaviour & Data Quality Audit

Business Intelligence Report · Rapid Scale SaaS · Prepared for Marketing & Onboarding Teams

1. Introduction

Every month, the Business Intelligence team reviews recent customer sign-up data ahead of the Monthly Business Review. This report audits the latest sign-up dataset (515 records from January–March 2025), fixes the data quality problems we found, and summarises what the cleaned data tells us about how customers are signing up, which plans they choose, and how that connects to early support activity. Our approach was to load and inspect the raw data, document and correct every quality issue with a clear rule for each fix, then re-run the standard sign-up analyses Marketing and Onboarding rely on every month.

2. Data Cleaning Summary

We started with 515 raw sign-up rows and worked through each issue in turn, documenting our reasoning so the fixes can be checked or repeated next month:

- **Duplicates:** 15 rows were exact or near-exact repeats of another customer_id. We kept the most recent record for each and removed the older duplicate, leaving 500 unique customers.
- **Plan & gender spelling:** plan_selected and gender both had inconsistent capitalisation and typos (e.g. “PRO”, “Proe”, “basci”; “MALE”, “Woman”, “NB”). We standardised both to a first-letter-capitalised form (Basic/Pro/Premium and Female/Male/Other), mapping known typos to their intended category.
- **Missing region (8.2%) and gender (4.3%):** both are categorical fields we cannot safely guess, so missing entries were labelled “Unknown” rather than dropped or imputed. This keeps the rest of each customer’s data usable while flagging the gap for Marketing to fix at the source.
- **Missing/invalid email (5.0%):** 25 blank addresses plus one literal “not-an-email” entry were flagged as invalid and left blank — we do not fabricate contact details.
- **Age (5.8% missing, plus 2 implausible values of 0 and 7 years old):** the two implausible ages were treated as data-entry errors and nulled out. All missing ages (32 rows total, 6.4%) were then imputed using the column median (33.5), which is robust to outliers and preserves sample size for age-based analysis, at the cost of slightly understating true age variation.
- **Out-of-range dates:** 2 rows were dated August 2026, 18 months outside the rest of the dataset. We flagged these as likely data-entry errors and excluded them only from the time-series (weekly trend) chart, keeping the rest of the row intact for all other analyses.

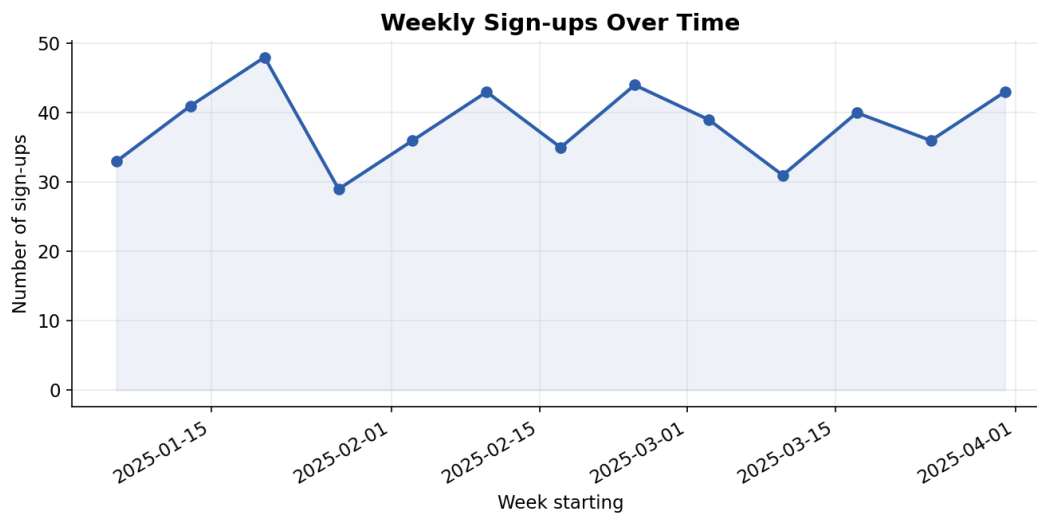
Data Quality Audit Table (prioritised by severity)

Issue	Scale	Business impact	Severity
Duplicate customer_id rows	15 rows (3.0%)	Inflates sign-up counts; double-counts customers in acquisition reporting.	High

Issue	Scale	Business impact	Severity
Missing region	42 rows (8.2%)	Regional campaign reports undercount / misattribute sign-ups.	High
Inconsistent plan_selected values	12 raw values → 3 categories	"Most popular plan" analysis unreliable without cleaning.	High
Missing age	30 rows (5.8%)	Understates sample size for age-based segmentation.	Medium
Missing / malformed email	26 rows (5.0%)	Customers unreachable by email marketing / onboarding.	Medium
Inconsistent gender values	11 raw values → 3 categories	Opt-in-by-gender analysis unreliable without cleaning.	Medium
Missing gender	22 rows (4.3%)	Slightly understates demographic breakdowns.	Low
Implausible age (<13)	2 rows (0.4%)	Would distort age statistics if left uncorrected.	Low
Out-of-range signup_date	2 rows (0.4%)	Would create a false spike in trend charts.	Low

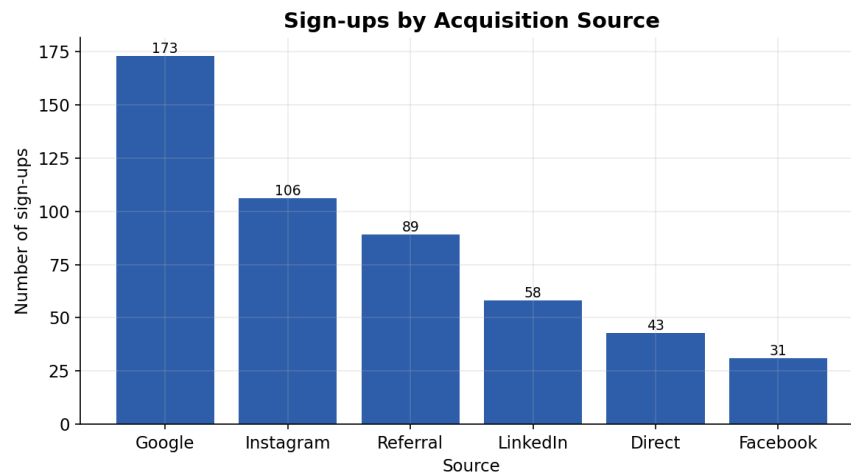
3. Key Findings & Trends

Sign-ups have held a steady, healthy pace through Q1, running between 29 and 48 per week with no sustained decline — a good baseline for planning Q2 acquisition targets.



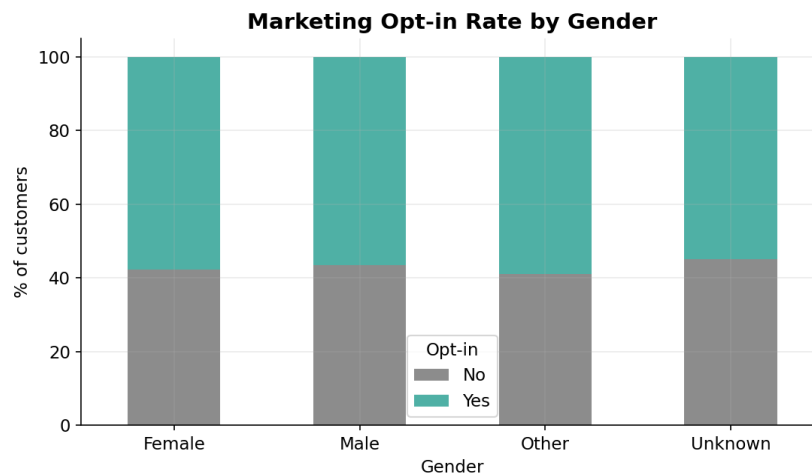
Weekly sign-up volume, Jan–Mar 2025 (2 out-of-range August 2026 rows excluded).

Google is the dominant acquisition channel, bringing in 173 of 500 customers (35%), more than Instagram and Referral combined for that share.



Sign-ups by acquisition source.

Marketing opt-in is consistent across gender groups, sitting between 55% and 59% for every group — gender is not a useful lever for improving opt-in rates.



Marketing opt-in rate by gender.

4. Business Question Answers

Which acquisition source brought in the most users last month?

Google, with 49 sign-ups in March 2025 — ahead of Instagram (38) and Referral (31).

Which region shows signs of missing or incomplete data?

Region itself is the main gap: 8.2% of all rows (42 of 515) have no region recorded. Missingness in other fields (email, age, gender) is fairly even across the known regions (roughly 3–9%), so this is a collection issue with the region field itself, not one particular region under-reporting.

Are older users more or less likely to opt in to marketing?

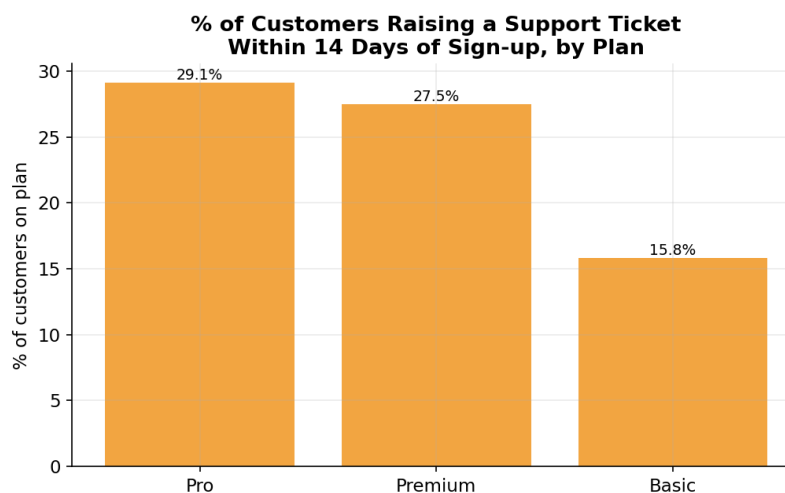
Barely at all — opt-in ranges narrowly from 47.8% to 61.4% across age groups, and the correlation between age and opt-in is essentially flat (-0.07). Age is not a meaningful predictor of opt-in behaviour.

Which plan is most commonly selected, and by which age group?

Basic is the most popular plan overall (190 of 500 customers), with Pro close behind (179). Basic leads among 18-25 and 46-65 year-olds, while Pro is the top choice for the core 26-45 age bracket.

Optional: Which plan's users are most likely to contact support? (Stretch task)

Joining support_tickets.csv to sign-ups shows **Pro** plan customers contact support earliest and most often: 29.1% raise a ticket within 14 days of signing up, vs. 27.5% for Premium and just 15.8% for Basic. Looking across the whole period, Premium has the highest share of customers ever raising a ticket (44.3%), Pro is close behind (40.2%), and Basic is notably lower (22.1%) — consistent with Basic being the simplest plan to onboard onto.



Share of customers raising a support ticket within 14 days of sign-up, by plan.

5. Recommendations

- **Double down on Google as the lead channel** and test whether the same creative/targeting approach can lift Instagram and Referral, which together still bring in fewer customers than Google alone.
- **Strengthen onboarding for Pro and Premium customers** in their first two weeks — both plans show meaningfully higher early support contact than Basic, suggesting more complex setup or expectations that a proactive welcome flow or checklist could ease.
- **Fix region capture at the point of sign-up** (e.g. make the field required, or infer it from IP/billing address as a fallback) so future regional reporting doesn't need a large "Unknown" bucket.

6. Data Issues or Risks

The highest-impact issue we found was **missing region data (8.2% of rows)**. Region underpins every territory-level report Marketing runs — campaign ROI by region, expansion planning, localisation decisions — and an 8% gap is large enough to shift which region looks like it's under- or over-performing, especially once regional numbers are split further by plan or source. Left uncorrected, this risks misdirected regional ad spend

or a missed expansion opportunity. The fix has two parts: going forward, make region a required field (or auto-populate it from billing/IP data) at sign-up; retrospectively, the 42 affected customers could be enriched later using billing address or support-ticket location data where available, rather than left as “Unknown” indefinitely.